

Ratio Lemma (abridged)

Ankit Bisain

Mop 2025

The full handout is at ankitbisain.github.io/files/RatioLemma.pdf.

§1 Theory

Here is a fact:

Fact 1 (Ratio Lemma). *Let ω be a circle through B and C , and let a line meet ω at points A, D and BC at E . Then,*

$$\frac{BE}{CE} = \frac{BA}{CA} \cdot \frac{BD}{CD}.$$

What this fact is actually saying is that if all of the points in your diagram are on a circle or a line, you understand collinearities very deeply. In particular, you can often solve a problem by just doing the following:

- Pick points X and Y such that most other points in the problem either lie on a circle through X, Y or on line XY . In a problem centered around $\triangle ABC$ and its circumcircle, this is often (but not always) B and C .
- Consider the function $\frac{XP}{YP}$ for points P either on a circle through X, Y or on a line through X, Y .
- Try to write the values of this function for many point in terms of a few (usually corresponding to the amount of points that can 'move' in the problem).

In essence, this is nice because it allows us to track a function of any point defined by either the intersection of a chord of a circle through X and Y with line XY , as well as intersections of similar lines with circles.

Note: to deal with config issues and make things easier to write, define $f(P) = \pm \frac{XP}{YP}$ to be negative when P is on segment XY or below line XY , and positive otherwise.

Here's an example, that I will probably do on the board or something (its written out in the original handout):

Example 1 (Pascal's). *Let A, B, C, D, E , and F be six points on a circle, and let XY be a chord of the circle. Suppose that AD and BC meet on XY , and CF and DE meet on XY . Prove that AF and BE meet on XY .*

Also turns out we also understand concyclicity very deeply:

Fact 2 (Condition for Circle and Line). *Let $ABCD$ be a cyclic quadrilateral, and let E and F be points on line BC . For all points Z , define $f(Z) = \pm \frac{BZ}{CZ}$ to be negative when Z is on segment BC or below line BC , and positive otherwise. Then, $ADEF$ is cyclic if and only if $f(A)f(D) = f(E)f(F)$.*

Fact 3 (Condition for Two Circles). *Let $ABCD$ and $BCEF$ also cyclic quadrilaterals with different circumcircles. For all points Z , define $f(Z) = \pm \frac{BZ}{CZ}$ to be negative when Z is on segment BC or below line BC , and positive otherwise. Then, $ADEF$ is cyclic if and only if $f(A)f(D) = f(E)f(F)$.*

Try proving that first theorem if you have time, it has some very instructive synthetic.
I will do this example on the board also

Example 2. *In triangle ABC , pick points D , E , and F on segments BC , CA , and AB , respectively, such that AD , BE , CF concur and $BCEF$ is cyclic. Prove that the circumcircle of DEF passes through the midpoint of BC .*

There's one more fact I should tell you

Fact 4 (Spiral Similarity). *Let ABC be a triangle, let E and F be points on AB and AC , and let $T \neq A$ be the intersection of the circumcircle of ABC and AEF . Then,*

$$f(T) = \frac{BE}{CF}.$$

Example 3 (Sharky Devil Point). *In triangle ABC with incenter I , the incircle touches BC at D . Point P on the circumcircle of ABC satisfies $\angle API = 90^\circ$. Prove that $\angle BPD = \angle CPD$.*

Oh and maybe also this

Fact 5 (Ptolemy). *For T on (ABC) , we have*

$$AT/CT \cdot BC + BT/CT \cdot CA + AB = 0,$$

with some choice of signs.

is good to know but it doesn't come up like super often.

§2 Problems

Have fun. Problems are ordered by length.

1. Distinct points B, C, X_1, X_2, Y_1, Y_2 lie on circle ω . Show that $\frac{BX_1}{CX_1} = \frac{BX_2}{CX_2}$ and $\frac{BY_1}{CY_1} = \frac{BY_2}{CY_2}$ if and only if $X_1Y_1 \cap X_2Y_2$ and $X_1Y_2 \cap X_2Y_1$ lie on BC .
2. Let ABC be a triangle, and let N satisfy $NBA \sim NAC$. Show that if the circle with diameter AN meets (ABC) at D and AC, AB at E, F , then AD, EF , and the tangent to (AN) at N concur.
3. (Butterfly Theorem) Let M be the midpoint of a chord PQ of a circle, through which two other chords AB and CD are drawn; AD and BC intersect chord PQ at X and Y correspondingly. Then M is the midpoint of XY .
4. Let AXY be an isosceles triangle, let M be the midpoint of XY , let A' be the reflection of A over XY , pick points B, C on XY , and let BA', CA' meet AX, AY at D, E . Then, show that $(AXY), (ABC), (ADE)$ are coaxial.
5. Let ABC be a triangle, D, E be points on BC , F be point on (ABC) , let (DEF) meet AD again at G and let (ADE) meet (ABC) again at H , and let (AFG) meet AB and AC at I and J . Then, show that AH, IJ , and GF concur.
6. Let $ABCD$ be a cyclic quadrilateral. Consider such pairs of points P, Q of diagonal AC that the rays BP and BQ are symmetric with respect the bisector of angle B . Prove that the center of the circumcircle of PDQ is on a fixed line.
7. Points D and E are selected on the circumcircle of $\triangle ABC$. AD and AE meet BC at X and Y . Let D' and E' be the reflections of D and E over the perpendicular bisector of BC . Prove that $D'Y$ and $E'X$ intersect on circumcircle of $\triangle ABC$.
8. Let ABC be a triangle with orthocenter H . Let AH, BH, CH , meet the opposite sides at D, E, F , and let D' be the reflection of D over the midpoint of BC . Show that if a point P on EF satisfies $AP \parallel BC$, then HP and AD' intersect on the circumcircle of AEF .
9. Let AA_1, CC_1 be the altitudes of triangle ABC , B_0 the common point of the altitude from B and the circumcircle of ABC ; and Q the common point of the circumcircles of ABC and $A_1C_1B_0$, distinct from B_0 . Prove that BQ is the symmedian of ABC (that is, $\frac{AQ}{CQ} = \frac{AB}{CB}$).
10. Let ABC be a triangle with circumcircle ω , let E, F be points on segments AC and AB , and let M be the foot of the altitude from the circumcenter of ABC to EF . Let AM meet ω at $D \neq A$, and suppose that EF, BC , and the tangent to ω at D concur. Show that M is the midpoint of EF .
11. Let $BCDEF$ be the vertices of a cyclic pentagon in some order. Suppose DE and DF meet BC at X and Y , and let the circumcircle of DEX meet the circumcircle of the pentagon at Z . Show that E, F , the reflection of Z across BC , and the reflection of D across the midpoint of BC are concyclic.